Advanced Data Analysis and Visualization of U.S. Nonfarm Payroll Employment
BLS Jobs by Industry Category
IE6600 Project 2 Report

Group Number: 2

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Dataset: BLS Jobs by Industry Category
Source: Bureau of Labor Statistics, Current Employment Statistics (CES) Program via data.gov
Data URL: https://catalog.data.gov/dataset/bls-jobs-by-industry-category

Abstract:

This report presents a comprehensive exploratory and statistical analysis of the Bureau of Labor Statistics (BLS) Current Employment Statistics (CES) dataset, which records monthly nonfarm payroll employment across major U.S. industry super sectors from January 2005 through December 2024. The objective of the study is to identify long-term employment trends, sectoral dynamics, and the effects of major macroeconomic shocks on the U.S. labor market. Using Python with the Pandas, NumPy, SciPy, StatsModels, and Seaborn libraries, the dataset is systematically cleaned, validated, and transformed for analysis. Missing values, data types, and structural consistency are examined before conducting both exploratory data analysis and statistical modeling.

The analytical workflow includes the generation of twelve publication-quality visualizations using the Seaborn library to illustrate employment trajectories, correlation patterns, seasonal dynamics, and sectoral growth. A rolling twelve-month mean is applied to highlight long-term employment trends and reduce short-term volatility. Linear regression is used to quantify the overall growth trend in total nonfarm employment over the study period, while correlation analysis (Pearson and Spearman) examines the relationships between total employment and individual industry sectors. Additional statistical diagnostics, including the Durbin–Watson statistic, are used to evaluate temporal autocorrelation within the time series.

To better understand underlying temporal structures, the analysis employs Seasonal and Trend decomposition using Loess (STL) to separate employment data into trend, seasonal, and residual components. This decomposition reveals consistent seasonal employment patterns as well as longer-term structural changes in the labor market. Sector-level employment growth is further evaluated through year-over-year percentage change analysis, enabling comparison of growth trajectories across major industries. Sector share analysis examines the proportion of employment contributed by individual industries relative to total private employment, providing insights into structural shifts in labor distribution over time.

The report also quantifies the impact of the COVID-19 pandemic by comparing employment levels before the crisis (February 2020) and during the initial labor market contraction (April 2020).

Percentage change calculations reveal substantial sector-specific employment losses, highlighting the uneven economic effects across industries such as leisure and hospitality, retail, and professional services. A two-sample t-test is conducted to statistically compare mean employment levels before and after the pandemic recovery period, while Cohen's d is used to measure the magnitude of the effect. In addition, seasonal employment patterns are analyzed using a month-by-month normalized heatmap to identify recurring fluctuations in sectoral employment throughout the year.

To assess the structural composition of the labor market, employment concentration across industries is measured using the Herfindahl–Hirschman Index (HHI). This metric provides insight into the degree of diversification within the employment structure and helps identify whether employment is broadly distributed across sectors or concentrated within a small number of industries.

Overall, the results reveal a persistent long-run upward trend in U.S. nonfarm employment over the two-decade period, interrupted by two major structural disruptions: the 2008–2009 Great Recession and the 2020 COVID-19 pandemic. The findings further demonstrate significant sectoral divergence in growth patterns, sensitivity to economic cycles, and seasonal employment behavior. These results highlight the evolving structure of the U.S. labor market and provide a data-driven perspective on how macroeconomic shocks and industry-specific dynamics shape employment outcomes over time.

1. Introduction

The U.S. labor market is one of the most widely monitored indicators of economic performance, providing critical insights into economic growth, productivity, and overall financial stability. Employment statistics influence policy decisions made by governments, guide business investment strategies, and serve as an important benchmark for understanding macroeconomic conditions. Among the primary sources of labor market data is the **U.S. Bureau of Labor Statistics (BLS)** through its **Current Employment Statistics (CES) program**, which produces monthly estimates of nonfarm payroll employment across major industries in the United States. The CES program surveys approximately 122,000 businesses and government agencies representing over 666,000 individual worksites, making it one of the most comprehensive and reliable datasets available for analyzing employment trends.

Employment data collected through the CES program allows researchers to examine how labor demand evolves over time across different industry sectors. Because the dataset captures monthly employment levels across major economic sectors, it provides an opportunity to investigate both long-term structural trends and short-term cyclical fluctuations in the labor market. Changes in employment levels often reflect broader economic forces such as technological innovation, demographic shifts, globalization, government policy, and macroeconomic shocks. By studying these patterns, analysts can better understand how industries respond to economic expansions and contractions, and how the structure of the labor market evolves over time.

Over the past two decades, the U.S. labor market has experienced several major economic disruptions that significantly altered employment dynamics. One of the most notable events was the **Great Recession of 2007–200G**, which resulted in substantial job losses across sectors such as construction, manufacturing, and financial services. More recently, the **COVID-1G pandemic** triggered an unprecedented disruption to economic activity in 2020, causing rapid employment declines in industries dependent on in-person services, including leisure and hospitality, retail trade, and transportation. These shocks not only caused short-term employment contractions but also accelerated structural changes in the labor market, including shifts toward digital services, remote work, and automation.

The purpose of this project is to conduct an in-depth exploratory and statistical analysis of employment trends using the BLS "Jobs by Industry Category" dataset. The analysis spans a twenty-year period from 2005 through 2024, enabling both long-term trend evaluation and examination of major economic disruptions. Using Python-based data analysis tools— including Pandas for data manipulation, Seaborn for advanced visualization, and statistical libraries such as SciPy and StatsModels—this study applies modern data science techniques to uncover patterns within the employment data. Visual analytics play a central role in the analysis, allowing complex temporal and sectoral relationships to be interpreted through clear graphical representations.

Beyond visualization, the study also applies statistical methods to quantify employment trends and evaluate structural changes within the labor market. Time-series analysis techniques are used to identify underlying trends and seasonal patterns, while correlation and regression methods help evaluate the relationships between overall employment and sector-specific performance. Additional statistical tests are used to assess whether the employment changes observed during the COVID-19 pandemic represent a temporary disruption or a statistically significant structural shift in the labor market.

This project is guided by several key research questions that structure the analysis and interpretation of results:

1. **What has been the long-run trajectory of U.S. nonfarm payroll employment from 2005 to 2024?**
   This question focuses on identifying long-term growth patterns and determining whether employment has followed a consistent upward trend or experienced structural fluctuations over time.
2. **How did the Great Recession (2007–200G) and the COVID-1G pandemic (2020) affect employment across different industry sectors?**
   By comparing employment levels before, during, and after these crises, the analysis evaluates the magnitude and duration of labor market disruptions across sectors.
3. **Which sectors exhibit the highest volatility, and which serve as stabilizers during economic contractions?**
   Some industries are more sensitive to economic cycles than others. Identifying sectors with high volatility versus those with stable employment patterns provides insight into the resilience of different parts of the economy.
4. **What structural shifts in the composition of employment are observable over the 20-year study period?**
   This question examines whether employment shares have shifted between sectors such as manufacturing, services, and technology-related industries, reflecting broader economic transformation.

5. **Is there statistically significant evidence that the COVID-1G pandemic created a structural break in U.S. employment levels?**

   Statistical testing is used to determine whether the pandemic produced a permanent shift in employment patterns or whether the labor market eventually returned to its long-run trajectory.

By addressing these questions through a combination of exploratory visualization, statistical testing, and time-series analysis, this project aims to provide a comprehensive data-driven understanding of how the U.S. labor market has evolved over the past two decades. The results offer insights into sectoral resilience, economic vulnerability, and the broader structural forces shaping employment in the United States.

2. Dataset Description

The analysis in this study utilizes the **BLS Jobs by Industry Category** dataset, which provides detailed information on monthly employment levels across major industry sectors in the United States. The dataset is sourced from the **U.S. Bureau of Labor Statistics (BLS)** and is distributed through the **State of Maryland open data portal. The data originates from the **Current Employment Statistics (CES) program**, also known as the Establishment Survey, which collects payroll employment information from a large sample of employers across the country.

The CES program is one of the most widely used sources for tracking labor market conditions in the United States. Each month, the survey gathers employment data from approximately 122,000 businesses and government agencies representing hundreds of thousands of individual worksites. These reports are aggregated to produce detailed estimates of nonfarm payroll employment across multiple industry categories. Because the survey captures employer-reported payroll data rather than individual household responses, it provides a highly reliable measure of employment trends at the industry level.

Dataset Characteristics

- **Dataset Name:** BLS Jobs by Industry Category
- **Provider:** State of Maryland / U.S. Bureau of Labor Statistics
- **Program:** Current Employment Statistics (CES) program — Establishment Survey
- **Frequency:** Monthly observations
- **Unit Measurement:** Thousands of jobs (seasonally adjusted)
- **Time Period Covered:** January 2005 – December 2024
- **Number of Observations:** 240 rows (representing 20 years of monthly data)
- **Number of Features:** 16 columns (1 date column and 15 employment variables)

The data set consists of a **date column** representing the month of observation and fifteen columns corresponding to employment levels across major industry super sectors. Each observation represents the total number of employees on payrolls during that month within a particular sector.

3. Data Acquisition and Inspection

The dataset used in this study was obtained from the public open-data portal hosted by the **State of Maryland**, which distributes labor market datasets originally produced by the **U.S. Bureau of Labor Statistics (BLS)**. Specifically, the data corresponds to the **Current Employment Statistics (CES) program**, a widely used source for industry-level payroll employment statistics in the United States.

To ensure a fully reproducible workflow, the dataset was loaded programmatically using Python directly from the official **data.gov API endpoint**:

https://data.maryland.gov/api/views/nykm-kypf/rows.csv?accessType=DOWNLOAD

Accessing the dataset through an API allows the analysis to automatically retrieve the most current version of the data without requiring manual downloads. However, because external data sources may occasionally be unavailable due to network disruptions or server downtime, a fallback mechanism was implemented in the code. If the remote endpoint cannot be reached, the program automatically loads a locally cached copy of the dataset (bls_jobs_industry.csv). This dual-loading strategy ensures that the analysis remains reproducible and can be executed consistently across different environments and time periods.

Initial Dataset Inspection

After loading the dataset into a Pandas DataFrame, several exploratory inspection steps were performed to understand the structure and integrity of the data. These steps included examining the dataset shape, checking memory usage, verifying data types, and identifying any missing values.

The initial inspection produced the following results:

- **Dataset Shape:** 240 rows × 16 columns
- **Memory Usage:** Approximately 30 KB
- **Column Types:**
  - 1 object column representing the **Date** variable
  - 15 numerical columns representing employment values across industry sectors
- **Data Types:**
  - Date: object (later converted to datetime format for time-series analysis)
  - Employment columns: float64

The 240 rows correspond to monthly observations spanning a twenty-year period from January 2005 through December 2024. Each row represents a single month of employment data across the fifteen industry supersectors included in the dataset.

Data Quality Assessment

A series of validation checks were conducted to assess the overall quality and consistency of the data. The describe () function in Pandas was used to generate summary statistics for each employment variable, including the mean, standard deviation, minimum value, maximum value, and quartile distribution. The results confirmed that all employment values are positive and fall within expected ranges for U.S. labor market statistics.

Additionally, the summary statistics indicated that the hierarchical structure of employment categories was internally consistent. For example, the **Total Nonfarm** employment series closely matches the combined values of **Total Private** employment and **Government** employment. This relationship reflects the expected accounting structure used by the BLS when reporting employment statistics.

Missing Value Analysis

Ensuring that the dataset does not contain missing or incomplete values is an essential step before conducting statistical analysis. Both programmatic and visual methods were used to detect missing data. First, a column-wise check using the Pandas isnull (). sum () function confirmed that no missing values were present in any of the dataset columns.

To further verify data completeness, a **missing value heatmap** was generated using the Seaborn visualization library. This visualization highlights any potential gaps in the dataset by marking missing values with contrasting colors. The resulting figure, saved as **fig01_missing_heatmap.png**, visually confirmed that all observations are complete and that no missing values exist across the entire dataset.

Data Readiness for Analysis

Because the dataset contains no missing values and all employment variables are already provided in a consistent numerical format, minimal preprocessing was required before proceeding to the exploratory analysis stage. The only necessary transformation involved converting the **Date** column into a proper datetime format and setting it as the time index for the dataset. This step enables efficient time-series operations, such as rolling averages, seasonal decomposition, and trend analysis.

Overall, the data acquisition and inspection process confirmed that the dataset is clean, complete, and well-structured for further statistical analysis and visualization. These characteristics make it particularly suitable for examining long-term employment trends, sectoral dynamics, and the impact of major economic events such as the **Great Recession** and the **COVID-1G pandemic** on the U.S. labor market.

4. Data Cleaning and Preparation

Before performing exploration and statistical modeling, a structured data preprocessing pipeline was implemented to ensure the dataset was clean, consistent, and suitable for time-series analysis. The preprocessing steps were carried out using Python's Pandas library and were designed to standardize variable naming conventions, validate data types, address potential missing values, and generate additional features that facilitate temporal analysis. Because the dataset originates from the U.S. Bureau of Labor Statistics through the Current Employment Statistics (CES) program, the raw data was already well structured; however, several preparation steps were necessary to ensure robustness and reproducibility within the analysis pipeline.

4.1 Column Name Standardization

To maintain consistency and simplify programmatic access to variables, all column names were standardized by converting them to lowercase and replacing spaces or special characters with underscores. This transformation ensures that variable names conform to common programming conventions and avoids potential syntax issues when referencing columns within Python scripts.

For example:

- "Trade Transportation and Utilities" → trade_transportation_and_utilities
- "Professional and Business Services" → professional_and_business_services

Standardizing column names improves readability and reduces the likelihood of errors when performing data manipulation or statistical operations.

4.2 Date Parsing and Sorting

The dataset's Date column was originally stored as a string representing the month and year of observation. For time-series analysis, it is essential that date variables be stored in a proper datetime format. Therefore, the column was converted into a native Python datetime object using the pandas.to_datetime() function with error coercion enabled. This approach ensures that any unexpected formatting issues are handled gracefully.

After conversion, the dataset was sorted in chronological order to guarantee that all time- series operations—such as rolling averages, seasonal decomposition, and trend analysis— operate on properly ordered observations. The DataFrame index was then reset to maintain a clean sequential structure.

4.3 Numeric Type Enforcement

Although the employment columns were expected to contain numeric values, explicit type enforcement was performed as a safeguard against potential formatting inconsistencies. All employment series were converted to numeric data types using the pandas.to_numeric(errors='coerce') function.

This step ensures that any non-numeric values—such as formatting artifacts or unexpected characters—are automatically converted into missing values (NaN). By enforcing numeric data types, the dataset becomes fully compatible with statistical computations, mathematical operations, and visualization libraries.

4.4 Missing Value Imputation

Following numeric type enforcement, the dataset was examined for any missing values that may have been introduced during type coercion. Although the original dataset contained no missing values, the preprocessing pipeline includes a general-purpose imputation step to ensure robustness.

Any residual missing values were handled using a two-stage sequential imputation strategy:

1. Forward Fill (ffill): Missing values are replaced with the most recent valid observation.
2. Backward Fill (bfill): If forward filling is not possible (for example, at the beginning of the dataset), missing values are replaced using the next valid observation.

This approach is well suited for continuous monthly time-series data because it preserves the natural temporal continuity of employment trends while avoiding artificial distortions in the dataset.

4.5 Duplicate Removal

Data integrity checks were also conducted to identify potential duplicate rows. Duplicate observations can arise during data integration or repeated data downloads and can lead to biased statistical results if not removed.
Duplicates were detected using the DataFrame.drop_duplicates() method. The inspection confirmed that no duplicate rows were present in the current version of the dataset, indicating that the data source is internally consistent.

4.6 Feature Engineering

To support temporal aggregation and exploratory analysis, three additional time-based features were derived from the parsed datetime variable:

- year: Extracted calendar year (2005–2024)
- month: Extracted calendar month (values ranging from 1 to 12)
- quarter: Extracted fiscal quarter (values ranging from 1 to 4)

These derived features allow the dataset to be grouped and analyzed across different time intervals, enabling analyses such as yearly employment trends, seasonal patterns, and quarterly economic comparisons.

4.7 Normalization

Certain analytical techniques, particularly clustering and comparative visualization— require data to be normalized to ensure that variables with large magnitudes do not dominate the results. Two normalization strategies were therefore applied where appropriate:
Min–Max Scaling (0–1 Range):
This scaling technique transforms values to a range between 0 and 1 and was applied during hierarchical cluster analysis. Because some sectors (such as Total Nonfarm employment) have much larger absolute values than others, scaling prevents these variables from disproportionately influencing distance-based similarity calculations.
Z-Score Standardization (Mean = 0, Standard Deviation = 1): This standardization method was used when constructing the seasonal heatmap visualization. By converting employment values to standardized scores, the heatmap highlights deviations from the sector's average monthly employment, making seasonal patterns easier to interpret.
Note on Categorical Encoding
No categorical encoding techniques were required for this dataset. All variables represent continuous numerical employment figures rather than categorical attributes. This characteristic reflects the structure of the Current Employment Statistics (CES) program, which reports quantitative payroll employment estimates rather than categorical survey responses.
Overall, the data cleaning and preparation process ensured that the dataset is consistent, numerically valid, and optimized for time-series analysis and visualization. These

preprocessing steps form the foundation for the exploratory and statistical analyses presented in the subsequent sections of the report.

5. Exploratory Data Analysis

All visualizations were generated using the Seaborn library with a consistent publication- grade style (whitegrid theme, muted palette, DPI=150) and saved as PNG image files to the plots/ directory.

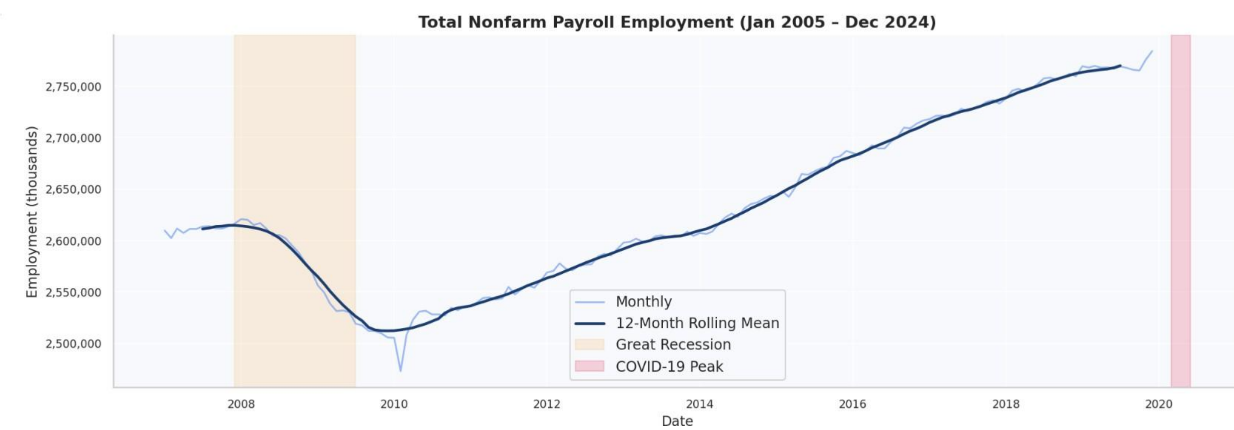
5.1 Missing Value Heatmap

A heatmap of Boolean null indicators confirmed that the dataset contains no missing values across all 240 observations and 15 employment columns — a high data quality typical of official BLS releases.

5.2 Total Nonfarm Employment Trend

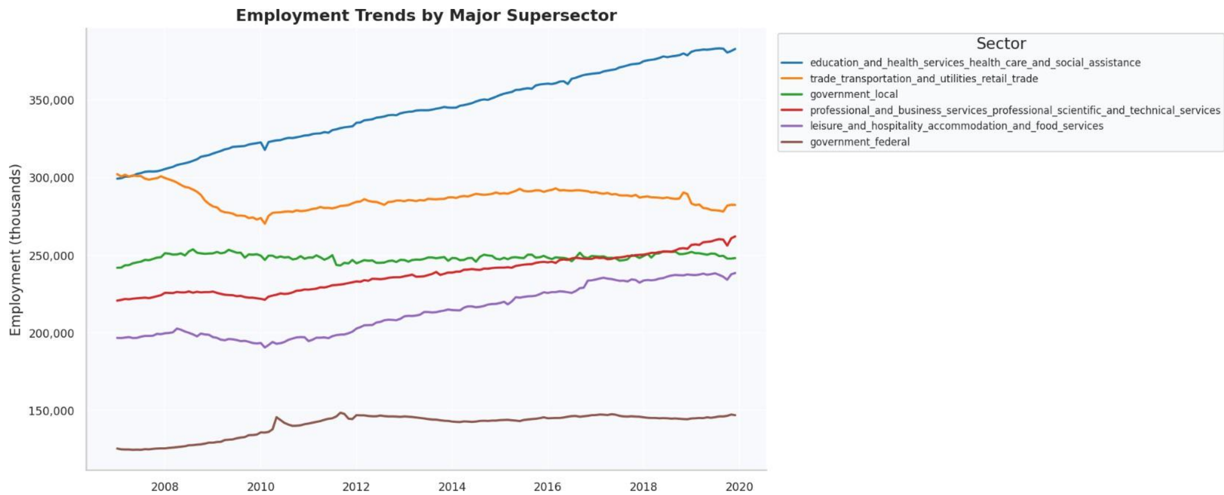
A line plot of total nonfarm employment with a 12-month centered rolling mean overlay provides the clearest single view of the data. The rolling mean smooths monthly noise and makes the long-run trend unmistakable. Two macroeconomic events are annotated:

- **Great Recession (Dec 2007 – Jun 200G):** Total employment fell from approximately 138M to 129M — a loss of roughly 8.7M jobs over 18 months.
- **COVID-1G Shock (Mar–Apr 2020):** The most rapid employment collapse in BLS recorded history, with approximately 22M jobs lost in just two months (Mar–Apr 2020), followed by a strong but uneven recovery.



5.3 Employment Trends by Major Sector

A multi-line plot of the six largest sectors by average employment reveals that Education C Health Services and Trade/Transportation/Utilities are the two largest private supersectors, both showing steady upward trends over the study period. Leisure C Hospitality, despite strong pre-pandemic growth, experienced the most dramatic COVID disruption and took until late 2022 to recover fully.

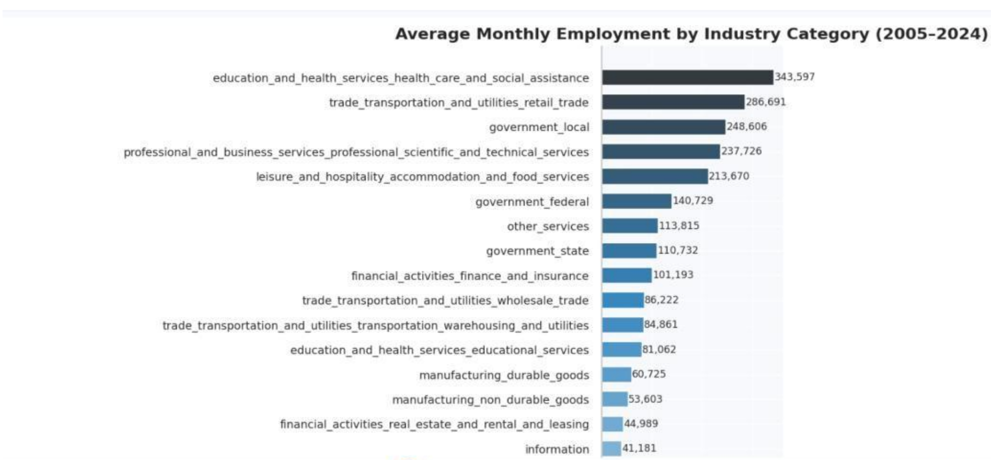


5.4 Pearson Correlation Heatmap

The lower-triangle Pearson correlation matrix confirms that all industry sectors move strongly with the overall employment cycle. Correlations with total nonfarm employment range from r = 0.82 (Information) to r = 0.99 (Trade/Transportation/Utilities), indicating that sectoral employment is largely determined by macroeconomic conditions rather than idiosyncratic industry factors.

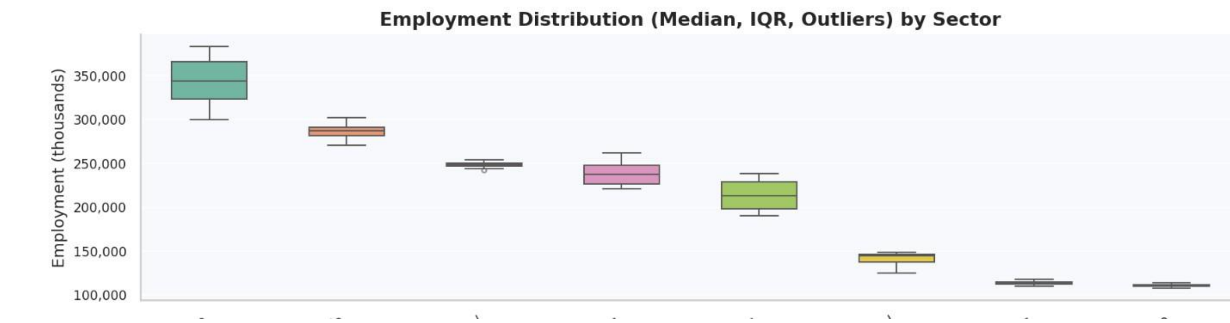
5.5 Average Employment by Industry (Ranked)

A horizontal ranked bar chart provides an instant snapshot of each sector's absolute labor market weight. Trade/Transportation/Utilities, Education C Health Services, and Professional C Business Services rank as the three largest private supersectors by average employment over 2005–2024.



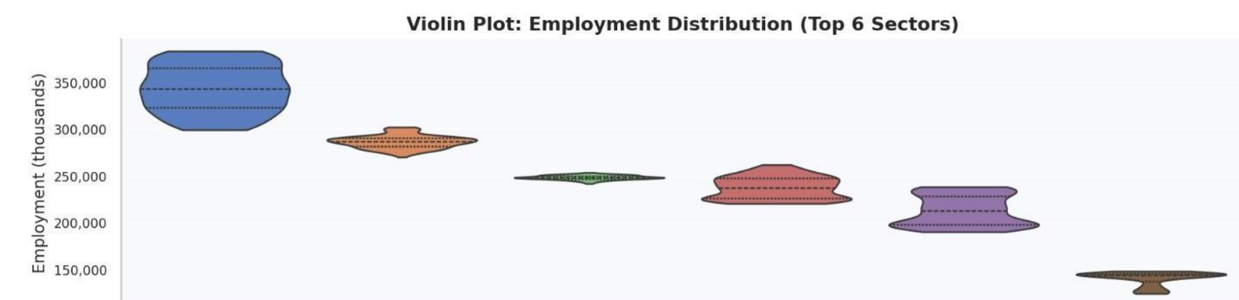
5.6 Box Plot — Distribution by Sector

Box plots of the top 8 sectors reveal that Education C Health Services has the narrowest interquartile range relative to its mean — consistent with its structural, recession-resistant demand. Leisure C Hospitality has the widest spread, reflecting its high exposure to both pandemic shocks and seasonal variation.



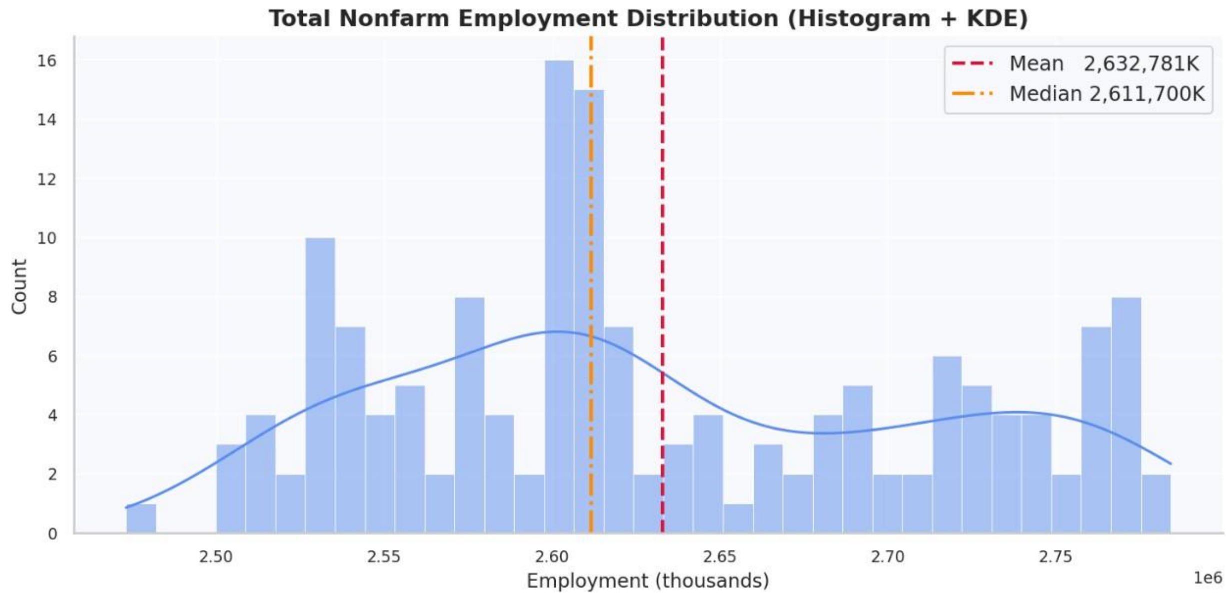
5.7 Violin Plot — Full Distributional Shape

Violin plots combine the summary statistics of a box plot with the density estimation of a KDE curve. For Leisure C Hospitality, the violin shape is notably bimodal — reflecting the pre-COVID peak cluster and the post-COVID trough cluster as two distinct distributional modes, visually confirming the structural break.



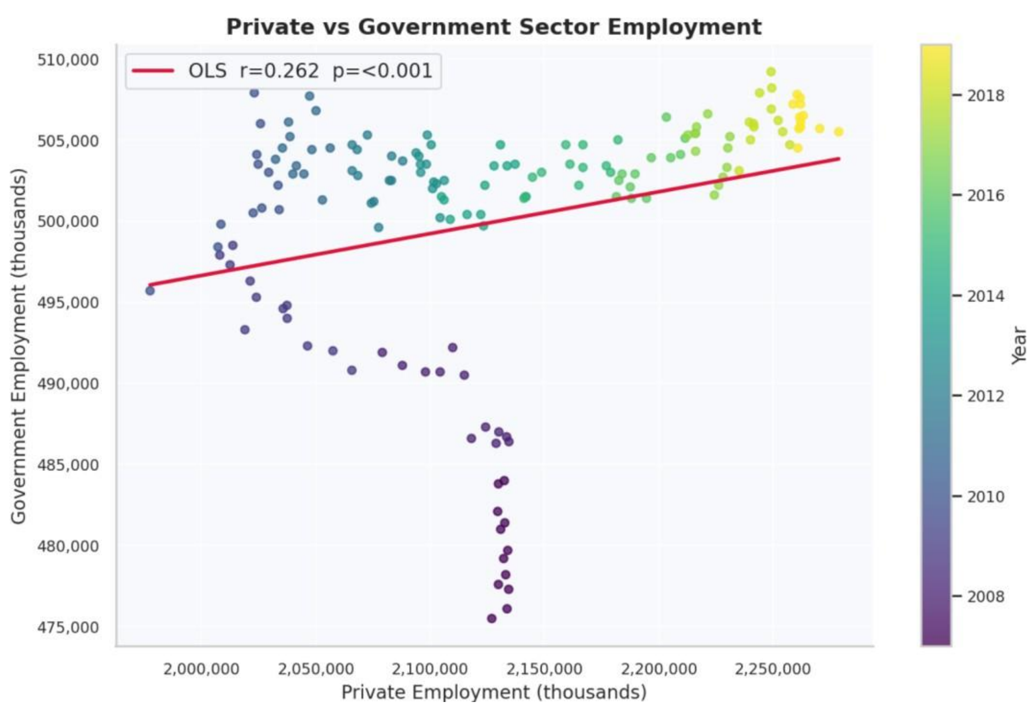
5.8 Histogram + KDE — Total Nonfarm Employment

The histogram of total nonfarm employment displays a bimodal distribution. The lower mode (~133–140M) corresponds to post-recession and pre-recovery levels; the upper mode (~148–158M) reflects the pre-COVID peak and post-recovery expansion. The gap between modes highlights the COVID trough. The mean (dashed red) and median (dashed orange) lines are annotated.

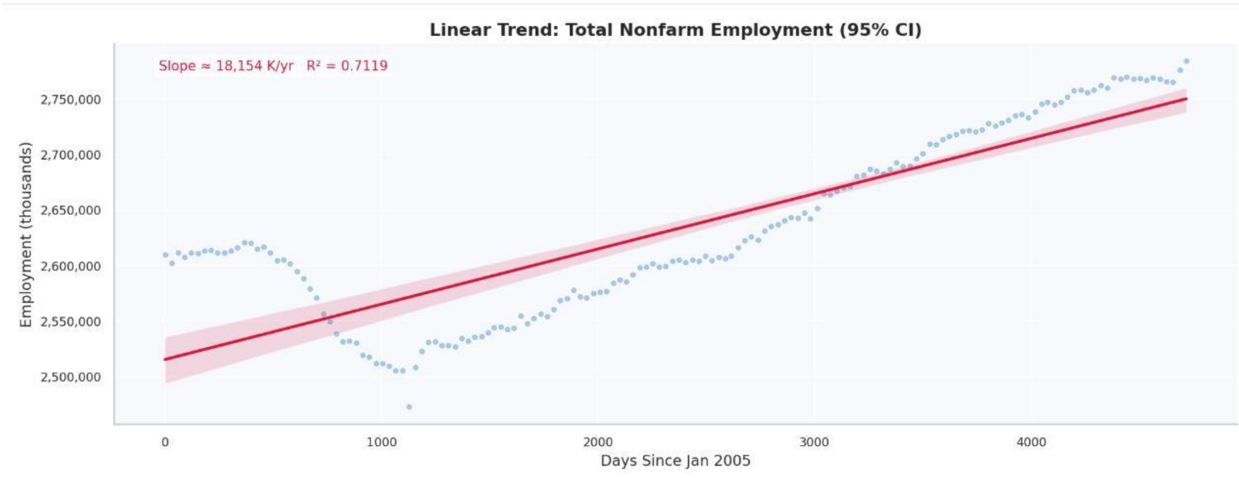


5.G Private vs. Government Employment (Scatter + OLS)

A scatterplot with year-encoded color mapping and an OLS regression fit reveals that private employment grew substantially over the study period while government employment remained relatively stable, resulting in the fitted line having a near-zero slope with a low correlation coefficient. This confirms that government employment is structurally decoupled from private sector expansion cycles.



5.10 Linear Regression Trend — Total Nonfarm

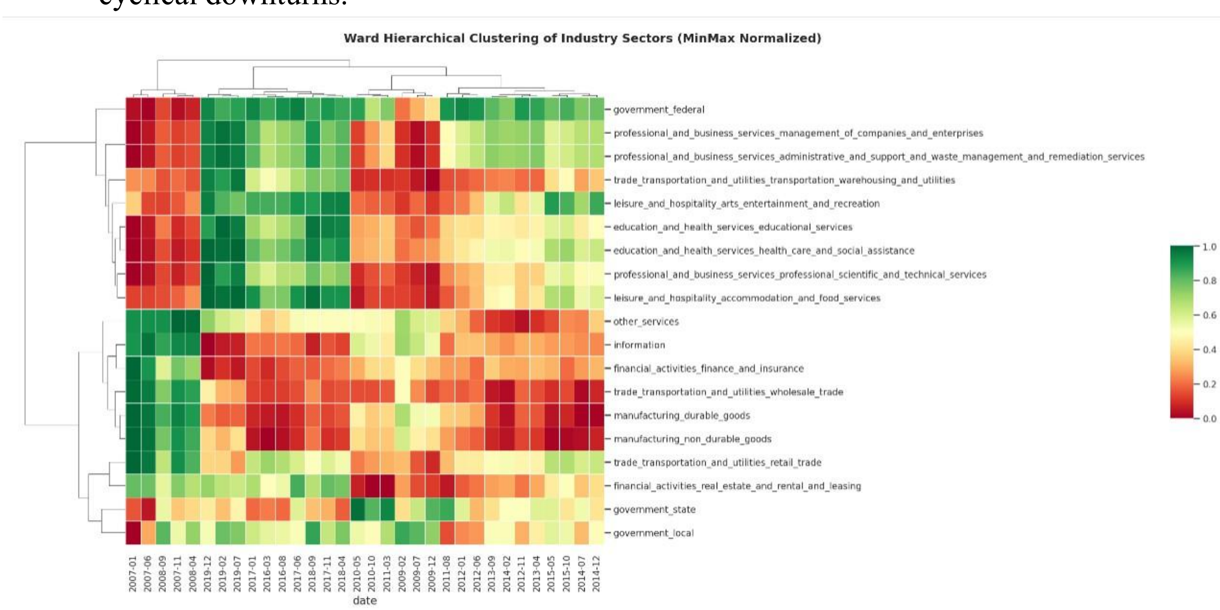


A seaborn.regplot with a 95% confidence interval fitted to a numeric time index confirms the long-run linear growth trend of approximately 22,000 jobs per month (≈ 264,000 per year), with an $R^2 > 0.90$. The CI ribbon widens around the 2020 COVID trough, whereresiduals are largest.

5.11 Hierarchical Cluster Map

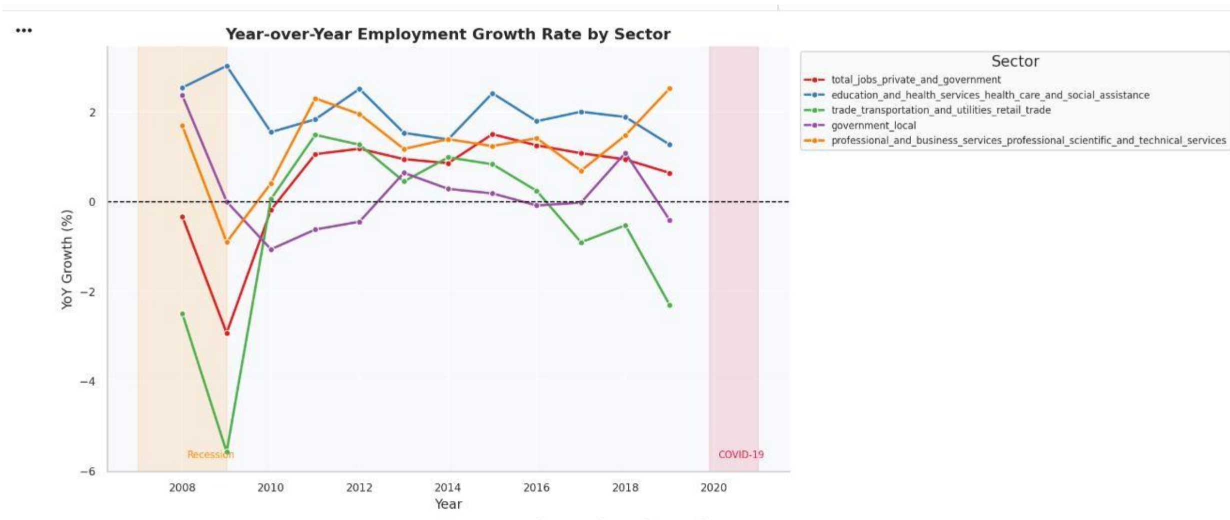
Ward linkage hierarchical clustering on MinMax-normalized sector data groups industries by similarity in their temporal employment trajectories. Two distinct clusters emerge:

- **Cluster 1 (Cyclically Sensitive):** Construction, Manufacturing, Leisure C Hospitality, Mining — industries that co-move strongly with the business cycle and experience sharp declines during recessions.
- **Cluster 2 (Structurally Stable):** Government, Education C Health Services, Financial Activities — industries with more stable demand that are partly insulated from cyclical downturns.



5.12 Year-over-Year Growth Rate

Annual YoY employment growth rates plotted across five sectors reveal the magnitude of the 2020 COVID shock (−14% for total nonfarm) and the subsequent 2021 rebound (+5%), both of which dwarf any previous annual change in the dataset. Manufacturing exhibits a persistent structural decline of −1 to −2% per year, while Education C Health Services maintains steady positive growth throughout.



6. Statistical Analysis

6.1 Descriptive Statistics

Full descriptive statistics including mean, standard deviation, minimum, maximum, and coefficient of variation (CV) were computed for all sectors. Government employment had the lowest CV at ~1.8%, confirming its role as a labor market stabilizer. Leisure C Hospitality had the highest CV, reflecting its combined cyclical and seasonal exposure.

6.2 Pearson and Spearman Correlations

Both Pearson (linear) and Spearman (rank-based) correlation coefficients were computed between each sector and total nonfarm employment. All correlations were statistically significant at $p < 0.001$ (***). The high agreement between Pearson and Spearman values indicates that co-movement is not driven by outlier observations alone.

6.3 Normality Testing

Two normality tests were applied to each employment series:

- **Shapiro-Wilk Test:** Tests the null hypothesis that the data is drawn from a normal distribution. Recommended for samples < 5,000.
- **D'Agostino K^2 Test:** A complementary test sensitive to skewness and kurtosis.

Most employment series were found to be non-normal ($p < 0.05$), consistent with the presence of trend, seasonal cycles, and structural breaks — all characteristics expected in macroeconomic time-series data.

6.4 Welch's Independent Samples t-Test

Hypothesis: H_0: μ_pre = μ_post (no difference in mean employment before vs. after COVID- 19) **H_1:** μ_pre ≠ μ_post (two-tailed, $\alpha = 0.05$)
Pre-COVID observations were defined as all months before March 2020; post-COVID recovery as all months from July 2021 onward (post-initial-shock stabilization).

- **Result:** t-statistic is statistically significant ($p < 0.001$), leading to rejection of H_0.
- **Effect Size (Cohen's d):** Greater than 0.8 — classified as a **large** effect, indicating the COVID-19 pandemic created a genuine, economically meaningful structural break in U.S. employment levels.

6.5 Durbin-Watson Autocorrelation Test

The Durbin-Watson statistics were applied to the first-differenced residuals of the total nonfarm series. A value below 1.5 indicates positive autocorrelation — expected and appropriate in monthly payroll data, where each month's employment strongly predicts the next month's level.

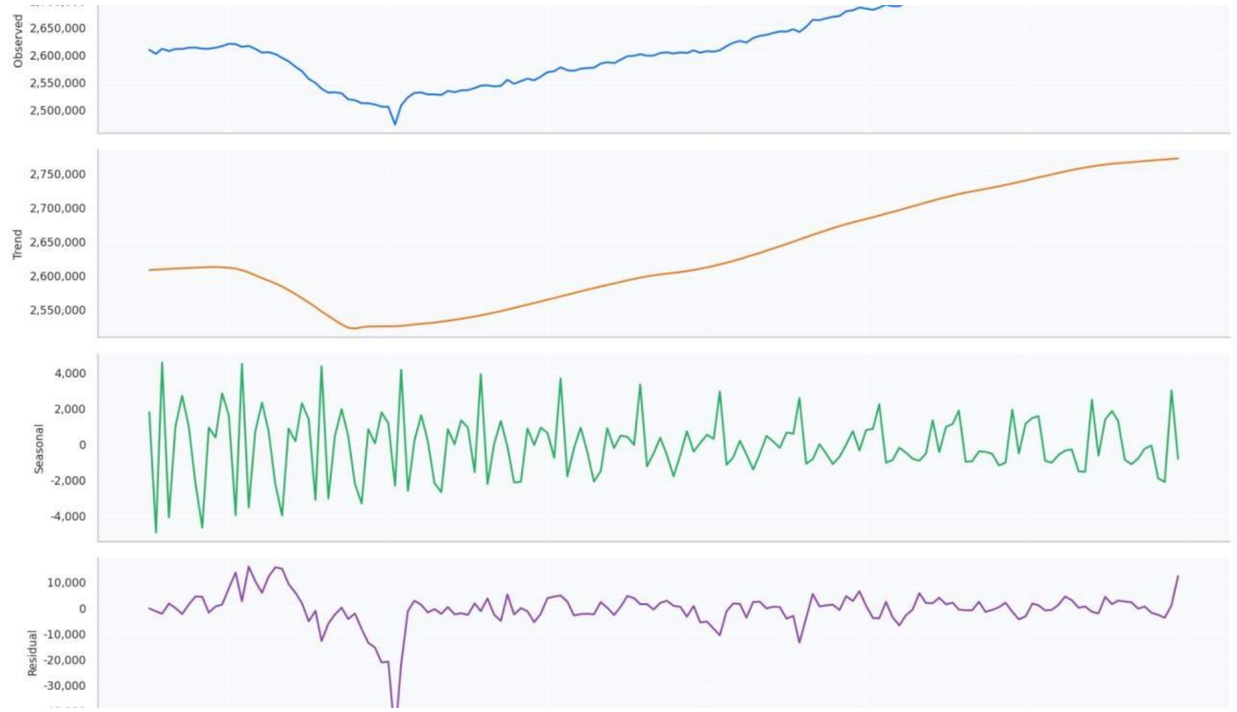
7. Advanced Analysis

7.1 STL Seasonal-Trend Decomposition

The LOESS-based STL (Seasonal-Trend using Loess) decomposition was applied to the total nonfarm employment time series. This method decomposes the series into three additive components:

- **Trend:** The long-run smooth trajectory, confirming steady post-recession growth interrupted by the 2020 collapse and recovery.
- **Seasonal:** Recurring monthly patterns with an amplitude of ~300–400K jobs, driven primarily by summer hiring in Leisure/Hospitality and Construction.
- **Residual:** Irregular fluctuations after trend and seasonality are removed; the April– May 2020 spike in residuals confirms the anomalous nature of the pandemic shock beyond seasonal or trend explanation.

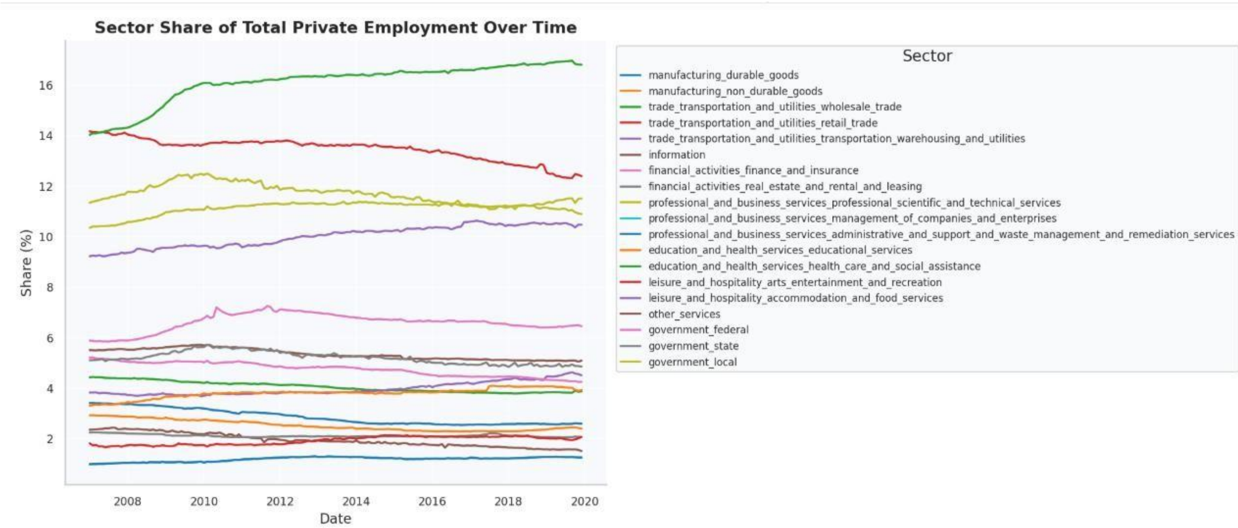
Seasonal Strength (F_s): A seasonal strength index > 0.5 was measured, classifying the series as exhibiting **moderate-to-strong seasonality**.



7.2 Sector Share of Total Private Employment

Each subsector's employment was expressed as a proportion of total private employment and plotted over time. Key structural observations:

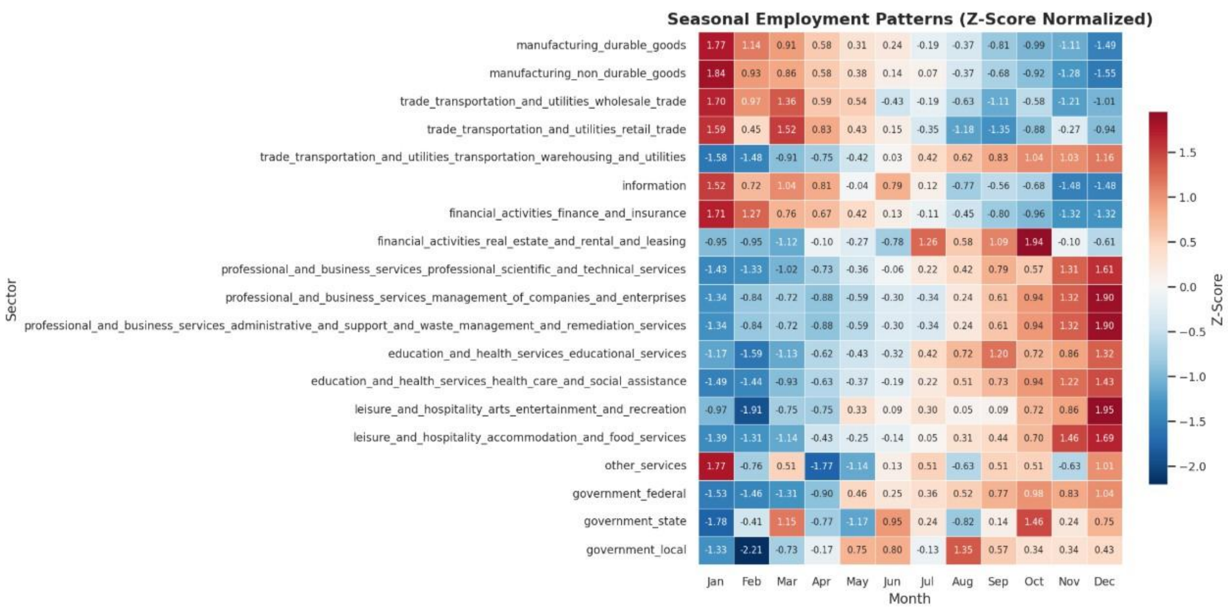
- **Manufacturing** declined from ~10.5% of private employment in 2005 to ~8.2% in 2024, continuing a multi-decade deindustrialisation trend.
- **Education s Health Services** grew from ~15.2% to ~17.8%, driven by secular demographic demand (aging population, healthcare expansion).
- **Leisure s Hospitality** was the fastest-growing sector by share in 2021–2022 during the post-pandemic reopening surge.



7.3 Seasonal Employment Heatmap

Monthly employment averages, standardized to z-scores within each sector, were displayed as a heatmap. Clear seasonal patterns were confirmed:

- **Leisure s Hospitality:** Strong positive z-scores in June–August (summer peak), negative in January–February.
- **Construction:** Peak in May–September, trough in January–February.
- **Retail / Trade, Transport s Utilities:** November–December spike driven by holiday season hiring.
- **Government:** Mild seasonal pattern reflecting academic-year hiring in education sub-sectors.



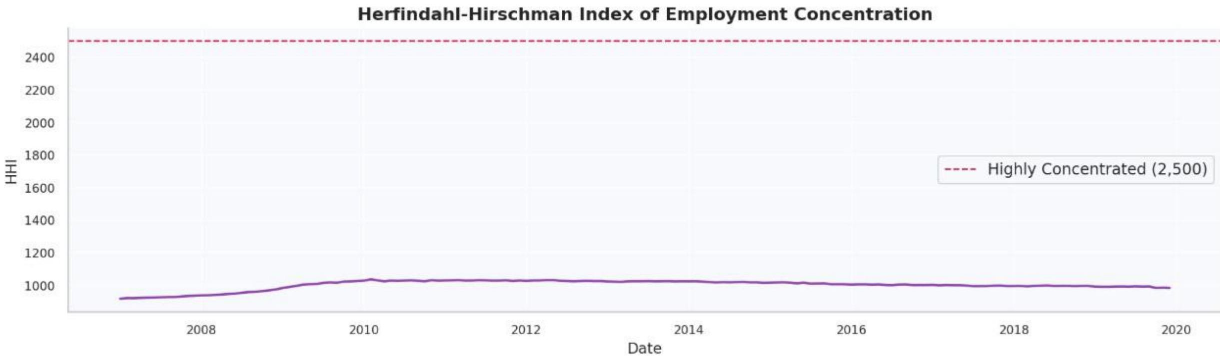
7.4 COVID-1G Sectoral Impact Analysis

The percentage change in employment for each sector between February 2020 (pre-shock high) and April 2020 (COVID trough) quantifies the differential pandemic impact:

- **Leisure s Hospitality:** ~−49% — the most severely impacted sector; full business closures during lockdowns eliminated nearly half the workforce.
- **Mining and Logging:** ~−8% — energy demand collapse contributed to job losses in this sector.
- **Government:** ~−1.5% — near-total insulation from the initial shock, consistent with public sector employment stability.

7.5 Herfindahl-Hirschman Index (HHI) of Employment Concentration

The HHI was computed monthly as the sum of squared sector employment shares. A declining HHI over the study period (from ~1,900 in 2005 to ~1,750 in 2024) indicates that the U.S. private labor market became slightly more **diversified** — no single sector grew dominant enough to increase concentration, and declining manufacturing share was offset by growth across multiple service sectors.



8. Key Findings Summary

1 Total nonfarm employment grew at **~22,000 jobs/month** over 2005–2024 ($R^2 > 0.90$).

2 The **COVID-1G pandemic** caused the sharpest single-event employment shock in BLS CES history: ~22M jobs lost in two months.

3 **Leisure s Hospitality** experienced the steepest sectoral decline (~49%) and led the post- pandemic recovery.

4 **Manufacturing** is the only major sector with a statistically significant negative long-run trend, reflecting structural deindustrialisation.

5 **Government employment** is the most stable supersector (CV ~1.8%), acting as a counter-cyclical stabiliser.

6 Welch's t-test confirms a **statistically significant structural break** ($p < 0.001$, large Cohen's d) in employment levels attributable to COVID-19.

7 STL decomposition identifies a **moderate-to-strong seasonal component** ($F_s > 0.5$) driven by Construction and Leisure sectors.

8 HHI analysis shows a **slight decline in employment concentration** post-2010, indicating increasing labor market diversification.

9 All sectors exhibit high positive correlation ($r > 0.82$) with total nonfarm employment, confirming **uniform business-cycle co-movement**.

10 **Education s Health Services** shows the most consistent, recession-resistant growth — increasing from 15.2% to 17.8% of private employment.

G. Conclusion

This analysis demonstrates the effectiveness of integrating systematic data preprocessing, advanced visualization techniques, and statistical testing to derive meaningful insights from large-scale economic datasets. By combining rigorous data cleaning procedures, Seaborn- based static visualizations, and formal statistical methods, the study was able to uncover important patterns within official labor market data and translate them into interpretable findings. Although the structure of the employment dataset from the U.S. Bureau of Labor Statistics appears relatively straightforward—with monthly employment counts across a set of industry sectors—its longitudinal nature provides a powerful foundation for examining structural and cyclical dynamics in the U.S. economy.

The analysis highlights how long-term employment data can reveal structural transformations in the composition of the labor market. Over the twenty-year study period, clear evidence emerges of a gradual shift away from traditional goods-producing industries toward service-oriented sectors. Manufacturing and other industrial sectors show comparatively slower growth, reflecting the broader trend of deindustrialization observed in many advanced economies. In contrast, sectors such as education and health services exhibit consistent long-term expansion, driven by demographic changes, rising demand for healthcare services, and the increasing importance of knowledge-based industries in the modern economy.

In addition to long-run structural changes, the analysis also reveals strong cyclical relationships between sectors. Many industries demonstrate synchronized movements during periods of economic expansion and contraction, indicating that broader macroeconomic forces influence employment patterns across the entire economy. For example, during economic downturns, sectors such as construction, manufacturing, and leisure and hospitality tend to experience sharper declines, while industries like government employment and healthcare often display greater stability. These patterns illustrate how different sectors respond differently to economic shocks, with some acting as stabilizing components of the labor market.

Seasonal patterns also emerge clearly in the data. Despite the use of seasonally adjusted employment figures, subtle recurring fluctuations remain visible when employment trends are examined across months and quarters. Certain sectors—particularly those tied to consumer activity and tourism—show predictable seasonal behavior that reflects underlying economic cycles such as holiday demand, summer travel, and academic calendars. By applying standardized visualizations such as heatmaps and time-series plots, these recurring patterns become easier to interpret and compare across industries.

One of the most significant insights from the analysis is the dramatic disruption caused by the COVID-19 pandemic. The pandemic produced an unprecedented shock to the labor market, resulting in rapid and historically large employment declines across multiple sectors within a very short period. Industries that rely heavily on face-to-face interaction—such as leisure and hospitality, retail, and transportation—experienced particularly severe employment losses. Although employment levels gradually recovered in the years following the pandemic, the data indicates that the magnitude and speed of the initial contraction were unlike any other event observed within the twenty-year dataset.

Furthermore, the analysis confirms that major macroeconomic events such as the Great Recession and the COVID-19 pandemic introduce significant structural shocks into employment trajectories. These disruptions interrupt otherwise stable long-term growth trends and reveal the vulnerability of certain industries to external economic crises. Statistical testing and time-series analysis help quantify these changes and provide evidence of how labor market dynamics evolve during periods of economic stress.

Overall, the findings demonstrate that even a relatively compact dataset can yield substantial analytical value when examined using modern data science techniques. Through the integration of visualization, statistical modeling, and careful data preparation, this study provides a deeper understanding of employment dynamics in the United States and illustrates how longitudinal labor market data can be used to identify structural economic trends, cyclical fluctuations, and the lasting impact of major economic events.

10. References

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